Dheeraj Kumar

Backend Software Engineer

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Professional Summary

A focused Backend Software Engineer with **2+ years of fintech experience** building high-throughput payment systems using **Java, Spring Boot, and Microservices**. Deep expertise in **REST API design, Oracle, Redis, IBM MQ, Apache Kafka** and database optimization with strong knowledge of **PCI-DSS** security standards. Experienced across the full delivery lifecycle in Agile teams. Solved **600+ DSA problems**; achieved Global Rank 206 in competitive programming.

Work Experience

PayNways India Pvt. Ltd. – Fintech Payments Platform Jan 2024 – Present
Software Engineer – Backend *Noida, India*

- **Technologies used:** Java, Spring Boot, Oracle, MSSQL, Hibernate/JPA, IBM MQ, Redis, AWS S3, GIAM, Swagger
- Designed and delivered **scalable RESTful payment APIs** using Java and Spring Boot, increasing transaction throughput by **25%** on a platform handling millions of daily financial operations.
- Architected a multi-database integration layer with **Oracle and MSSQL** via Hibernate ORM; optimized JPQL queries, maintaining **99.9% system uptime** across critical payment flows.
- **Performance Optimization:** Reduced API response latency by **30%** through Hibernate query tuning, Java multithreading, and asynchronous event processing via **IBM MQ**.
- Built a **KYC/AML compliance module** integrating third-party identity verification APIs with secure **AWS S3** document storage, enabling automated risk-flagging and improving onboarding by **30%**.
- Secured microservices with **certificate-based GIAM authentication** and fine-grained RBAC, eliminating unauthorized access across all enterprise service endpoints.
- Configured **stored procedures and SQL functions** for payment logic, loan/account ID generation, and API responses; integrated **Swagger (OpenAPI)** for complete auto-generated API documentation.
- Integrated **Apache Kafka** for asynchronous logging and **RestClient** for synchronous service communication; implemented structured logging with **Log4j2 and SLF4J**.
- Led resolution of **5+ critical production tickets/defects**, ensuring clear communication and timely resolution of project workflows and issues, **reducing error rates by 20%**.

Technical Skills

Primary Skills:	Java, Spring Boot, Spring Data JPA, Hibernate, Microservices, REST APIs, Oracle, Redis, MSSQL
Languages:	Java, Python, PL/SQL, SQL, JavaScript, HTML5, CSS3
Backend:	Spring MVC, Spring Security, JWT, Swagger/OpenAPI, RBAC, Kafka, IBM MQ, Multithreading
Databases:	Oracle, MSSQL, MySQL, MongoDB
Tools & Infra:	Git, Maven, Docker, Postman, IntelliJ IDEA, STS, AWS S3, GIAM, Log4j2, SLF4J, Linux
CS Fundamentals:	DSA, OOPs, PCI-DSS, System Design, Design Patterns, Agile/Scrum, Java Memory Model

Personal Projects

Digital Wallet GitHub Live *Java, Spring Boot, MySQL, Redis, JWT, REST APIs*

- Architected a secure digital wallet with **UPI P2P fund transfers**, wallet top-up, and real-time balance reads via **Redis cache** (5-min TTL) and full transaction history.
- Established **idempotent payment APIs** using Redis distributed key locking (24h TTL) to prevent duplicate transactions under high-concurrency retry scenarios.
- Enforced **RBAC** (ROLE_USER, ROLE_MERCHANT, ROLE_ADMIN) and **AES-256-GCM** encryption on all sensitive data; designed a **webhook reconciliation engine** (15-min cron), cutting discrepancies by **40%**.

High-Throughput UPI Payment Gateway GitHub Live *Java, Spring Boot, Kafka, Redis, MySQL, Microservices*

- Built a high-throughput UPI switch handling real-time payment routing, fund settlement, and inter-bank transaction management using **Apache Kafka** for async event streaming and **Redis** for distributed caching.
- Implemented **microservices architecture** with service discovery, circuit breakers, and centralized structured logging for fault tolerance and observability.

Education

Motilal Nehru National Institute of Technology – NIT Allahabad 2020 – 2023
Master of Computer Applications (MCA) *CGPA: **7.61 / 10.0*** *Allahabad, India*

Achievements & Competitive Programming

- Sharpened backend development skills by mastering **600+ DSA problems** across LeetCode and CodeChef.
- Selected as **Finalist** at **Codex**, the intercollegiate coding hackathon organized at **NIT Allahabad**.
- Achieved **Global Rank 206** in CodeChef Starters contest among 8,000+ competitive programmers.